

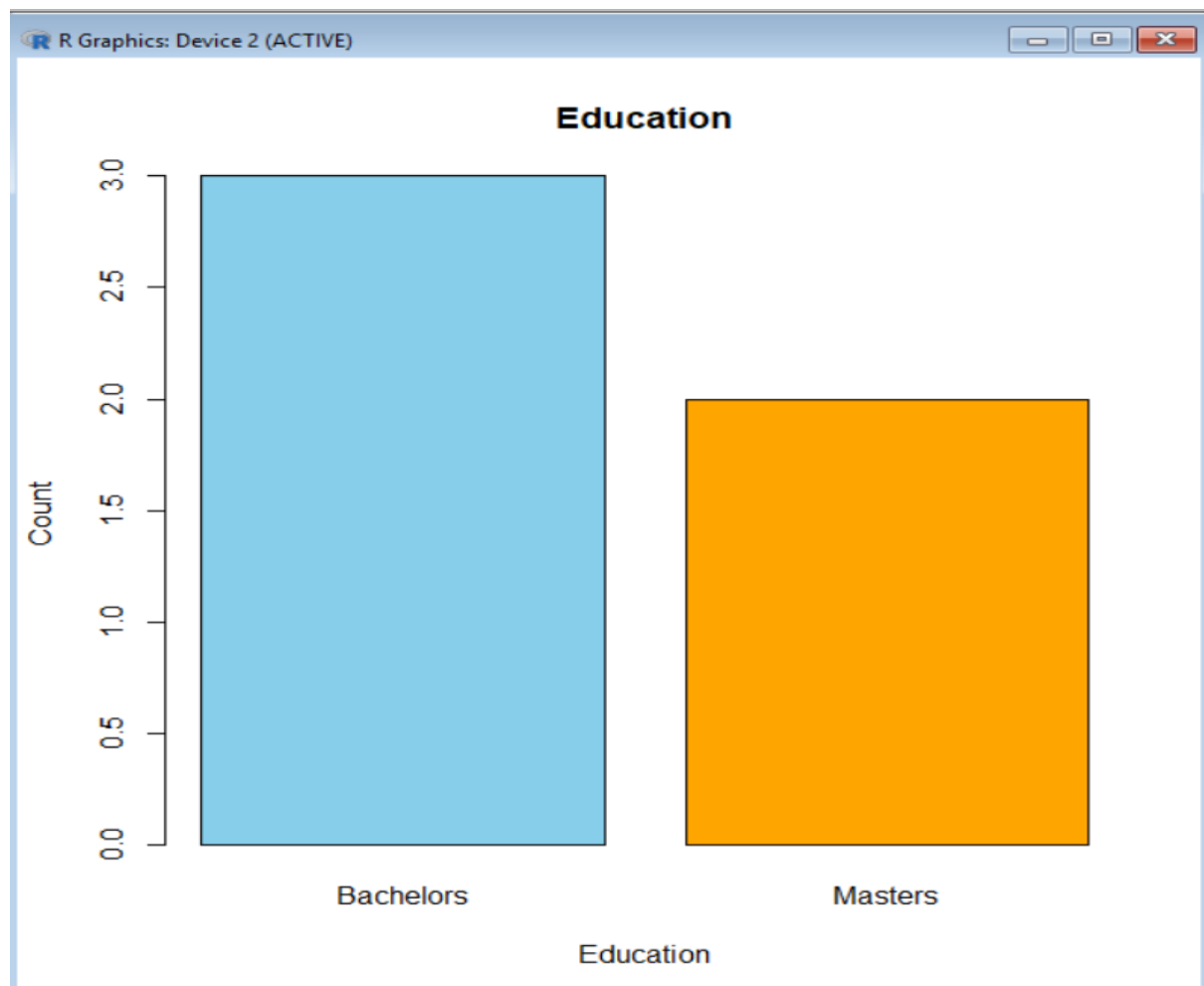
Name: B C Yugesh

Reg no: 192424282

Course Code: DSA0604

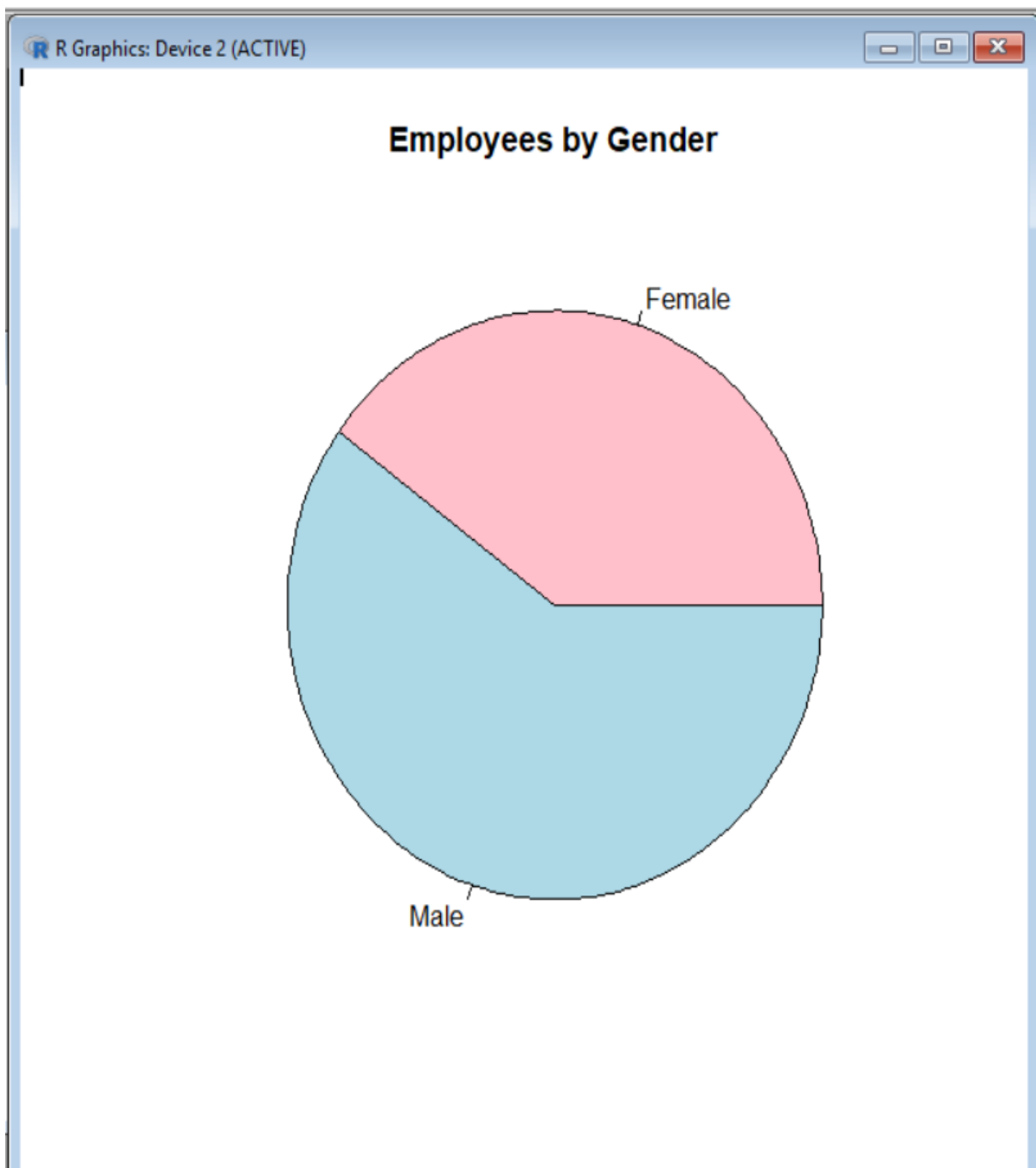
1. Bar Chart

```
employee <- read.csv("Employee.csv")  
edu <- table(employee$Education)  
barplot(edu,  
        col=c("skyblue","orange","green"),  
        main="Employees by Education",  
        xlab="Education",  
        ylab="Number of Employees")
```



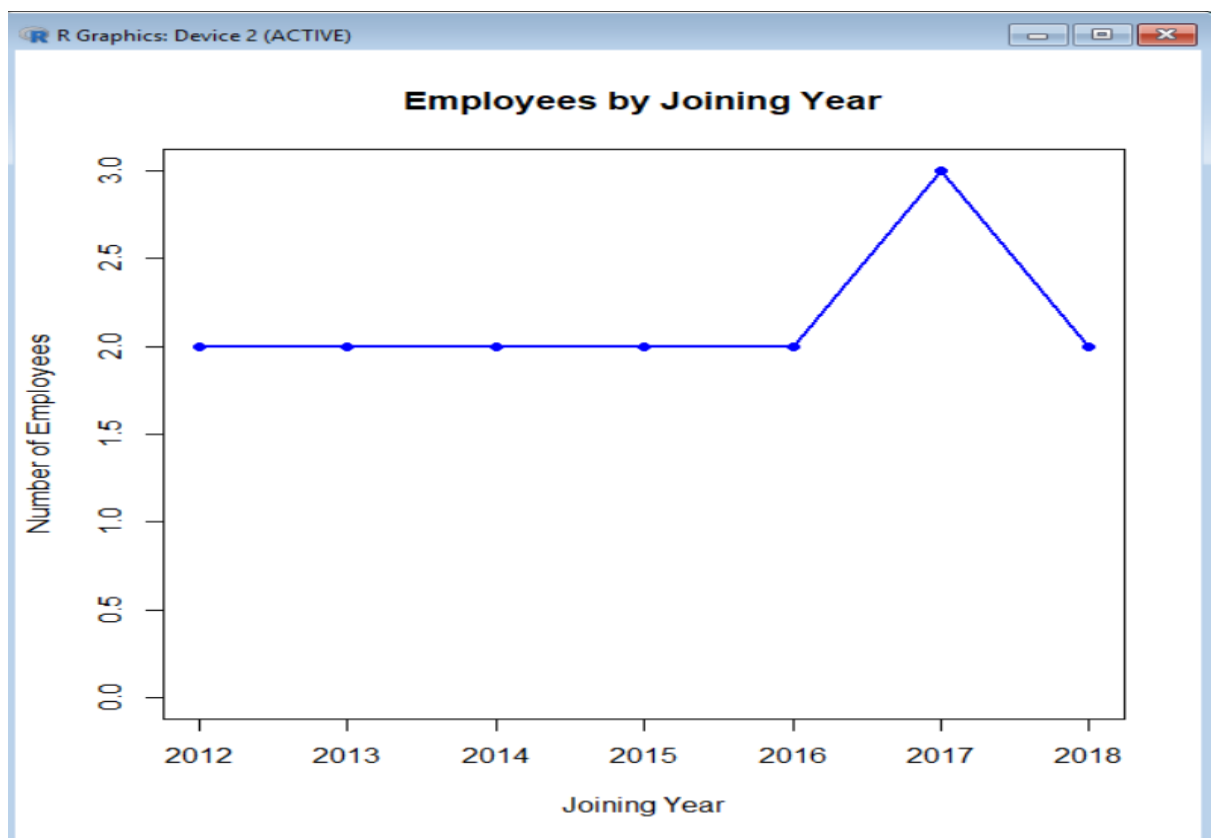
2. Pie Chart

```
employee <- read.csv("Employee.csv")  
gen <- table(employee$Gender)  
pie(gen,  
     col=c("pink","lightblue"),  
     main="Employees by Gender")
```



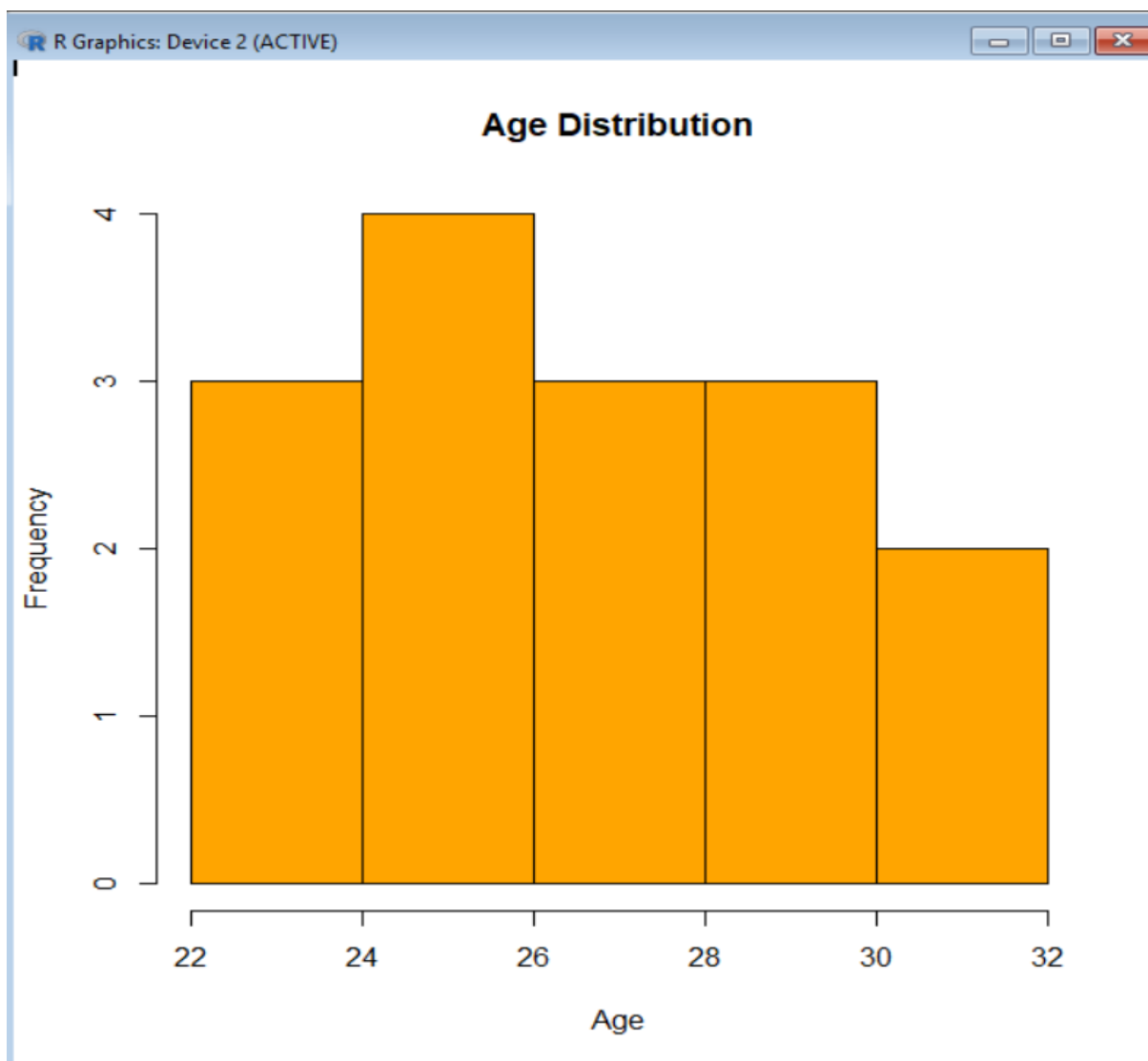
3. Line Chart

```
employee <- data.frame(  
  JoiningYear = c(2012,2013,2014,2015,2016,2017,2018,  
                  2016,2017,2015,2014,2013,2012,2018,2017)  
)  
join <- table(employee$JoiningYear)  
plot(join,  
      type="o",  
      col="blue",  
      lwd=2,  
      pch=16,  
      main="Employees by Joining Year",  
      xlab="Joining Year",  
      ylab="Number of Employees")
```



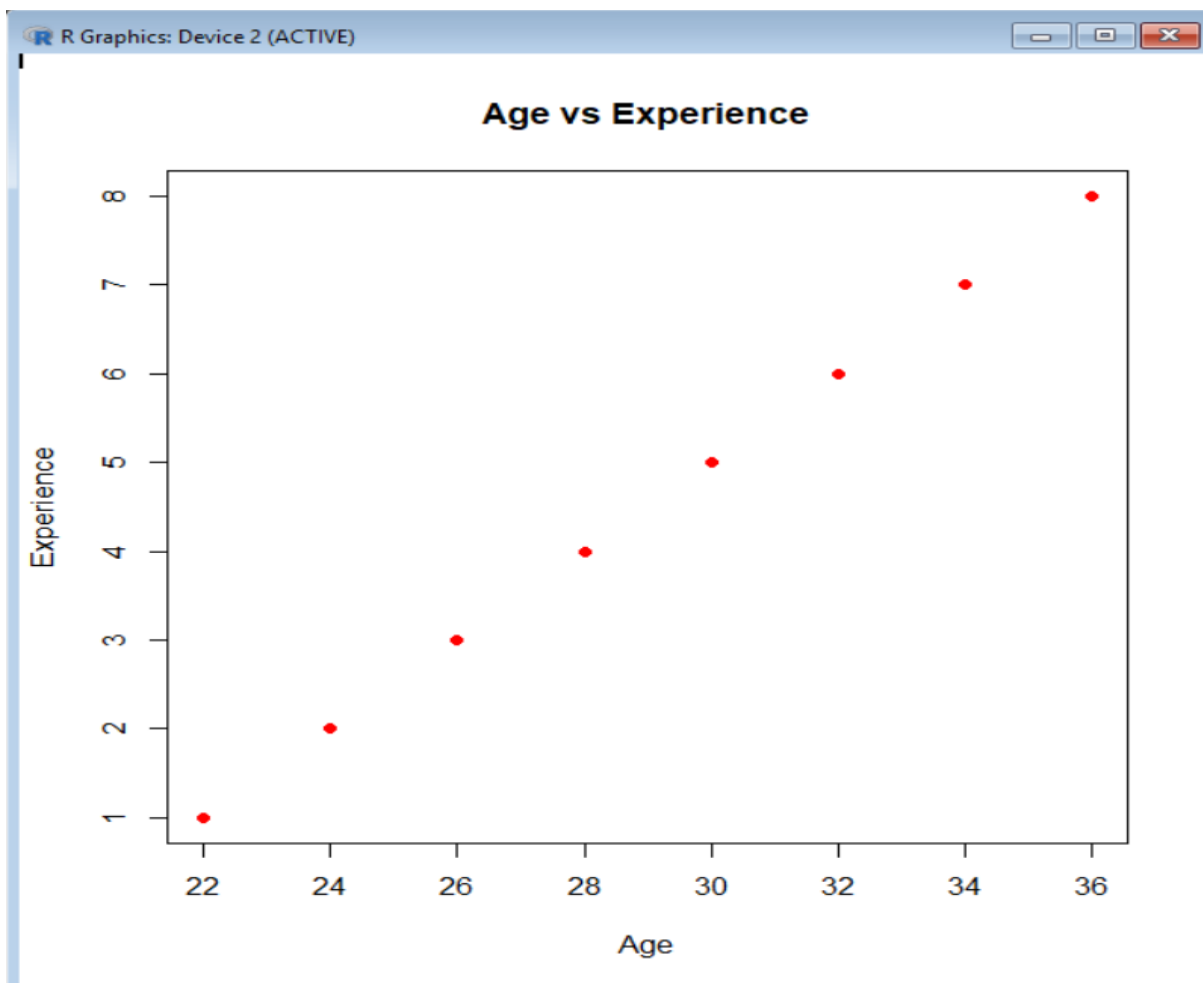
4. Histogram

```
employee <- data.frame(  
  Age=c(22,24,25,26,27,28,29,30,31,32,  
        24,25,26,28,30)  
)  
hist(employee$Age,  
      col="orange",  
      border="black",  
      main="Age Distribution",  
      xlab="Age")
```



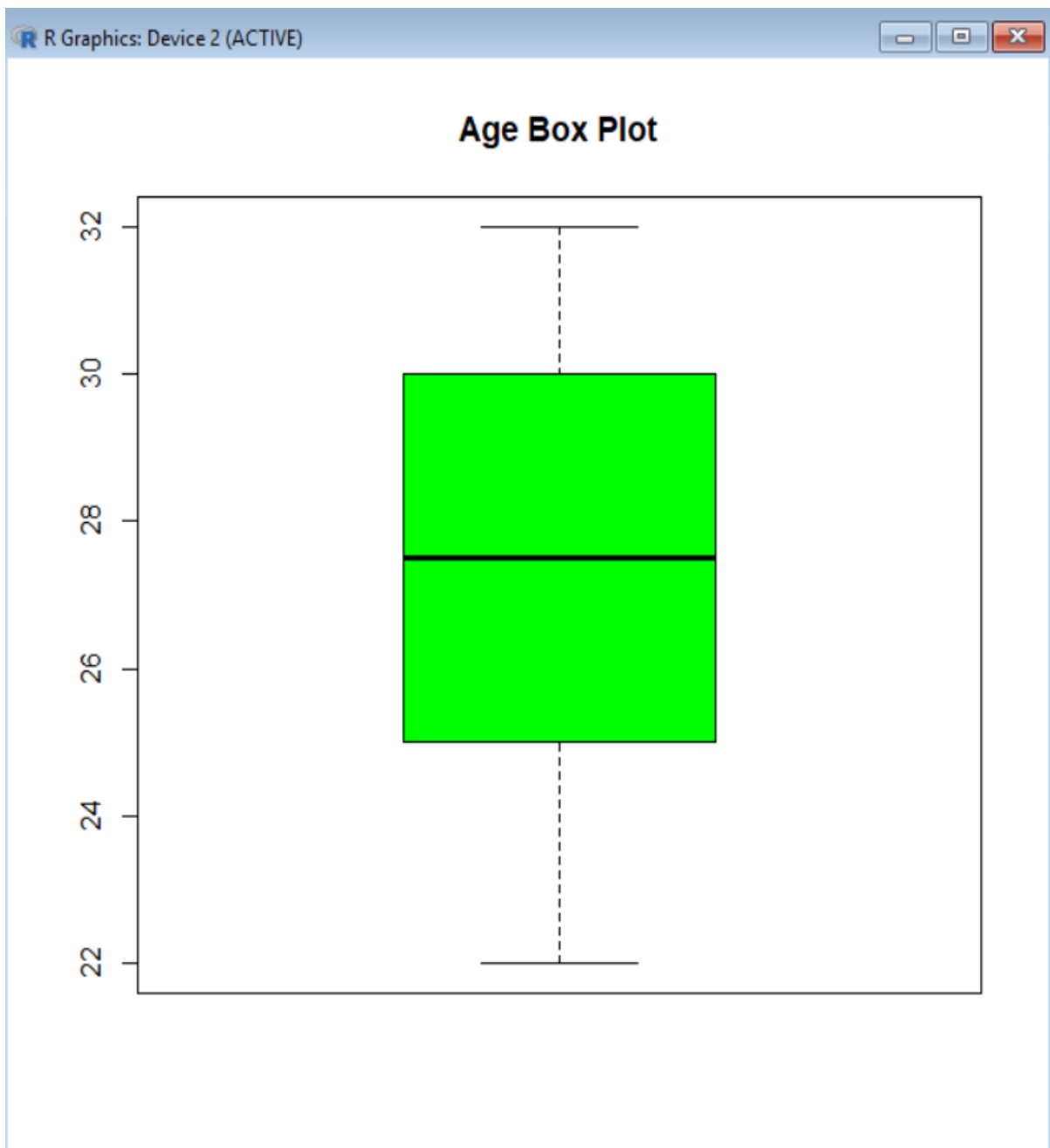
5. Scatter Plot

```
employee <- data.frame(  
  Age=c(22,24,26,28,30,32,34,36),  
  Experience=c(1,2,3,4,5,6,7,8)  
)  
plot(employee$Age,  
      employee$Experience,  
      col="red",  
      pch=19,  
      main="Age vs Experience",  
      xlab="Age",  
      ylab="Experience")
```



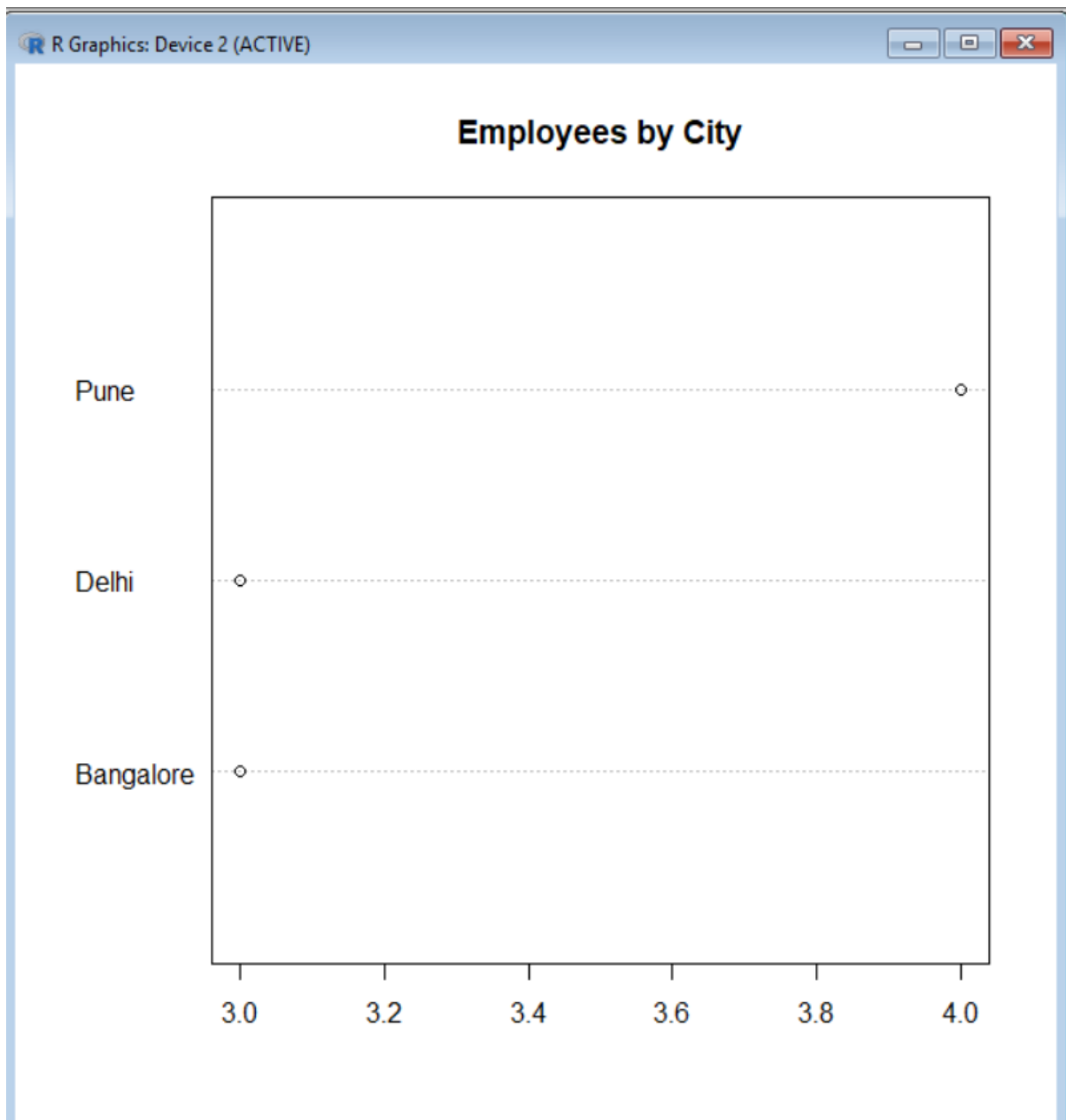
6. Box Plot

```
employee <- data.frame(  
  Age=c(22,24,25,26,27,28,29,30,31,32)  
)  
boxplot(employee$Age,  
  col="green",  
  main="Age Box Plot")
```



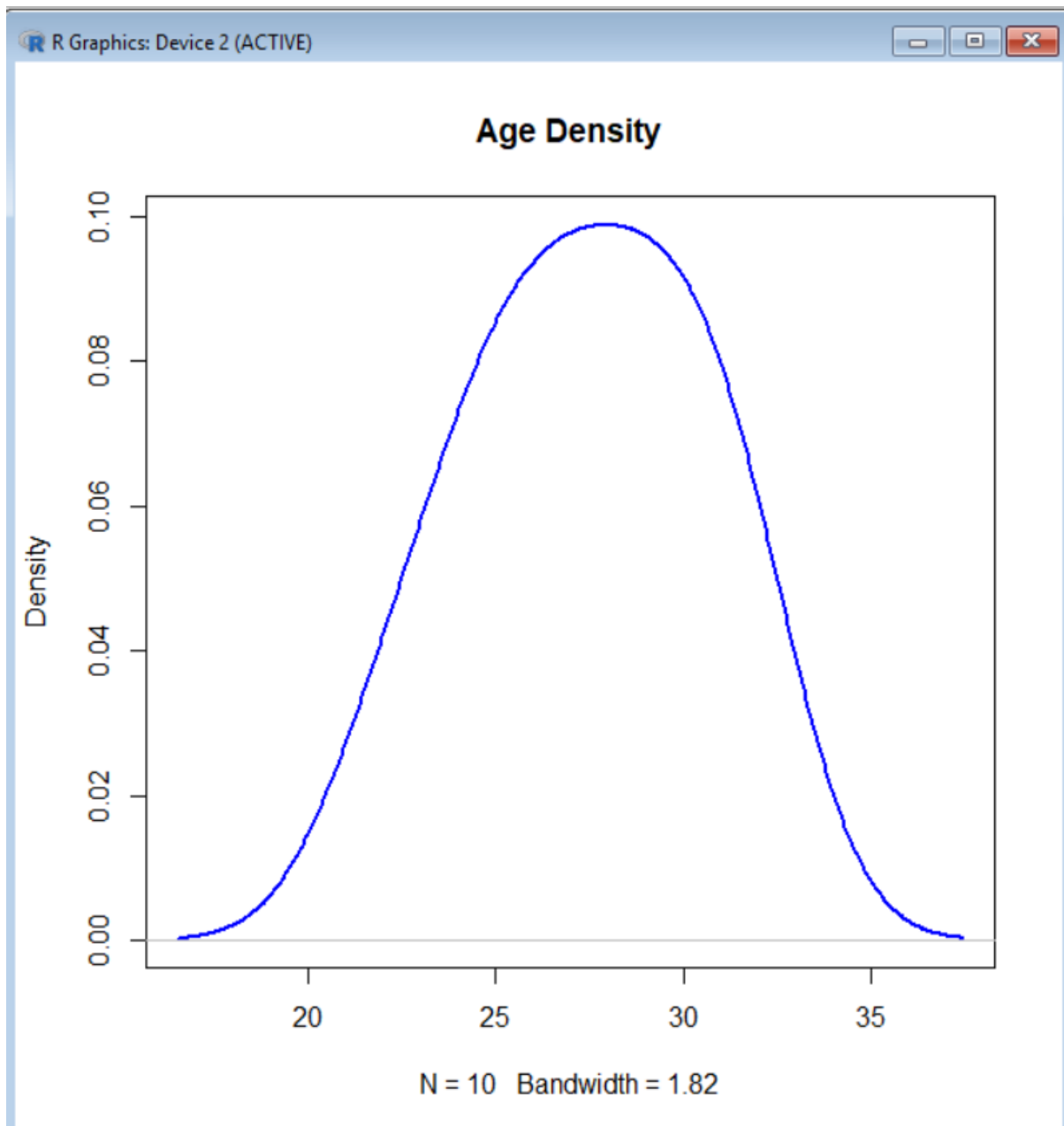
7. Dot Chart

```
employee <- data.frame(  
  City=c("Bangalore","Pune","Delhi","Pune",  
         "Delhi","Bangalore","Pune",  
         "Delhi","Bangalore","Pune")  
)  
dotchart(table(employee$City),  
         main="Employees by City")
```



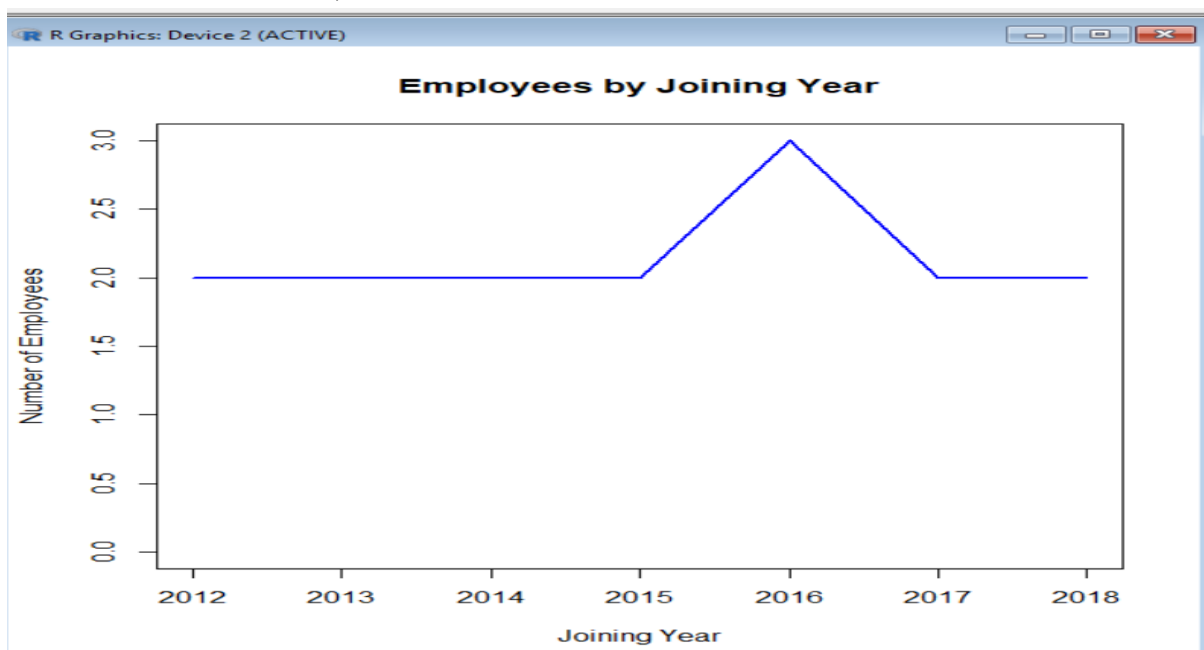
8. Density Plot

```
employee <- data.frame(  
  Age=c(22,24,25,26,27,28,29,30,31,32)  
)  
plot(density(employee$Age),  
     col="blue",  
     lwd=2,  
     main="Age Density")
```



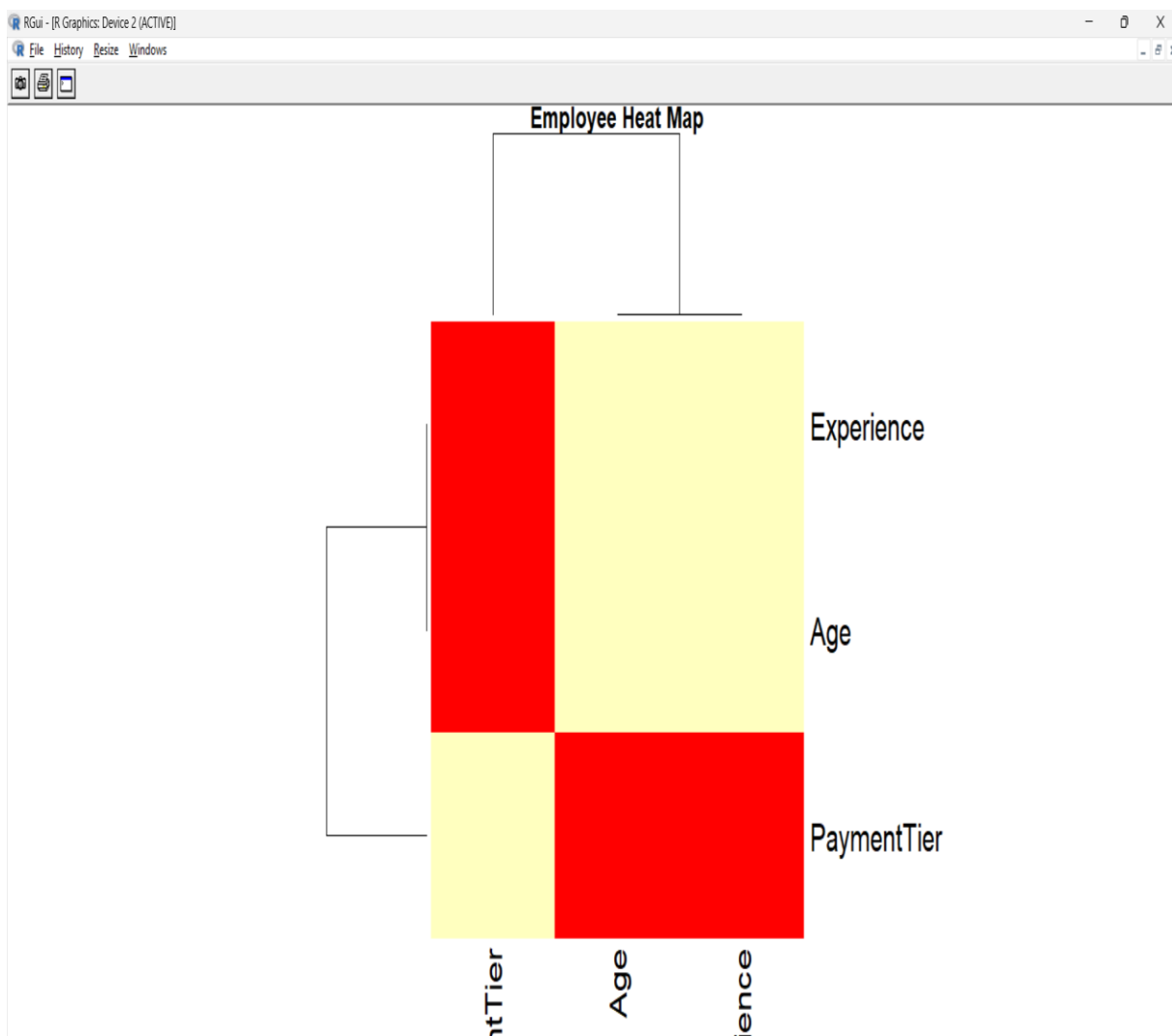
9. Area Chart

```
employee <- data.frame(  
  JoiningYear = c(2012,2013,2014,2015,2016,2017,2018,  
                  2016,2015,2014,2017,2018,2013,2012,2016)  
)  
join <- table(employee$JoiningYear)  
plot(join,  
      type = "l",  
      lwd = 2,  
      col = "blue",  
      main = "Employees by Joining Year",  
      xlab = "Joining Year",  
      ylab = "Number of Employees")  
polygon(c(1:length(join), length(join), 1),  
        c(join, 0, 0),  
        col = "lightblue",  
        border = "blue")
```



10. Heat Map

```
employee <- data.frame(  
  Age = c(22,24,26,28,30,32,34,36),  
  PaymentTier = c(1,2,2,3,3,2,1,3),  
  Experience = c(1,2,3,4,5,6,7,8)  
)  
data <- cor(employee)  
heatmap(data,  
  col = heat.colors(10),  
  scale = "none",  
  main = "Employee Heat Map")
```



11.Stem Plot

```
employee <- data.frame(  
  Age=c(22,24,25,26,27,28,29,30,31,32)  
)  
stem(employee$Age)
```

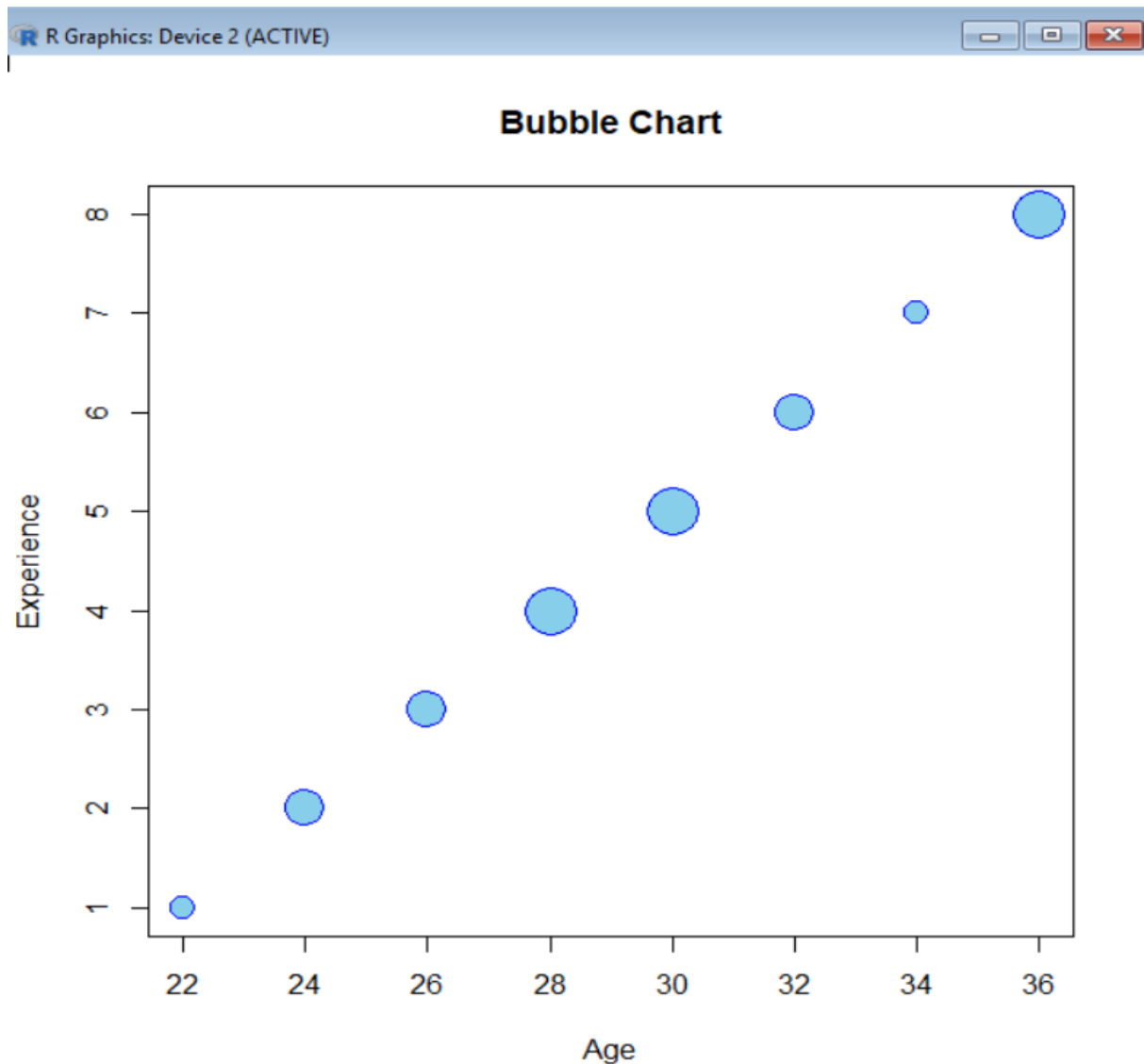
The decimal point is 1 digit(s) to the right of the |

```
2 | 24  
2 | 56789  
3 | 012
```

12. Bubble Chart

```
employee <- data.frame(  
  Age = c(22,24,26,28,30,32,34,36),  
  Experience = c(1,2,3,4,5,6,7,8),  
  PaymentTier = c(1,2,2,3,3,2,1,3)  
)  
plot(employee$Age,  
  employee$Experience  
  cex = employee$PaymentTier + 1,  
  pch = 21,  
  bg = "skyblue",  
  col = "blue",  
  xlab = "Age",
```

```
ylab = "Experience",  
main = "Bubble Chart")
```



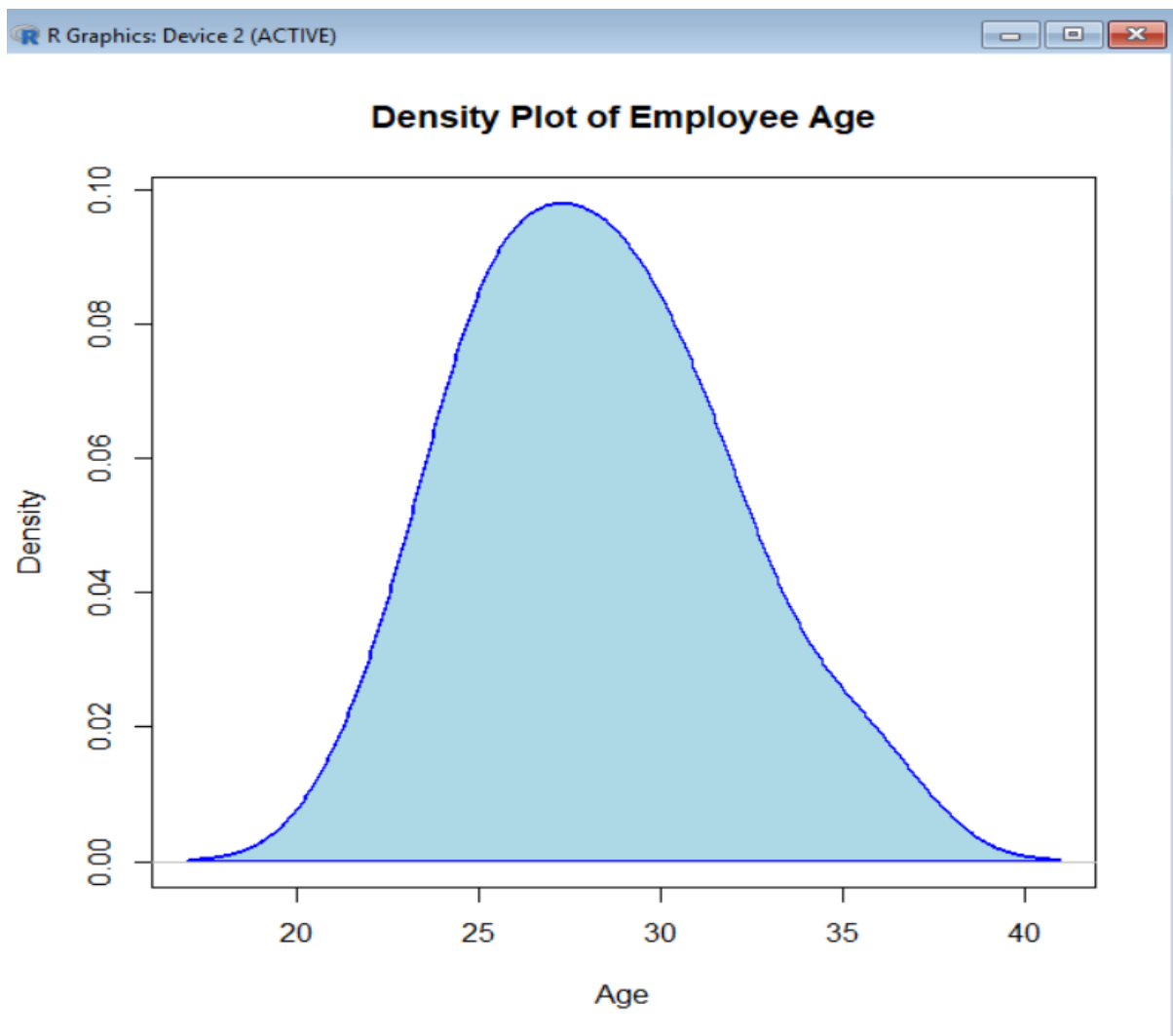
13. Density chart

```
employee <- data.frame(  
  Age = c(22,24,25,26,27,28,29,30,31,32,  
    24,25,26,28,30,32,34,36,27,29)  
)
```

```
# Density Plot
```

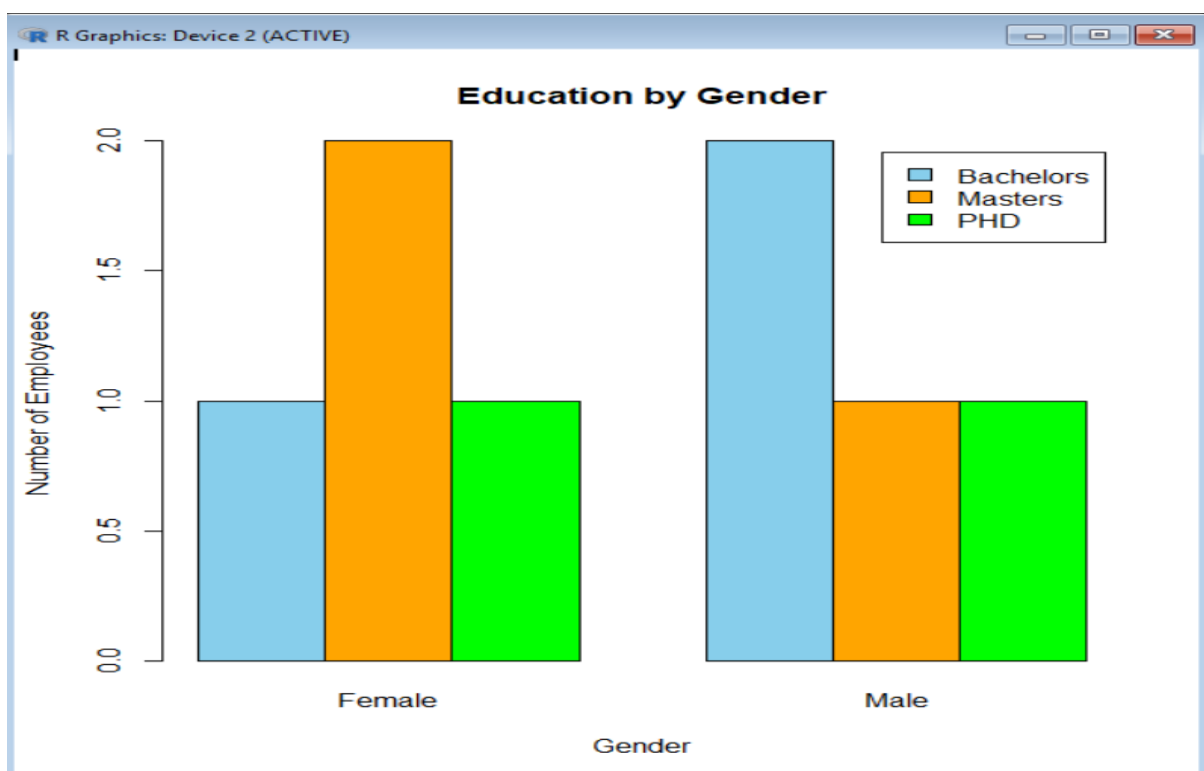
```
plot(density(employee$Age),  
     main = "Density Plot of Employee Age",  
     xlab = "Age",  
     ylab = "Density",  
     col = "blue",  
     lwd = 2)
```

```
polygon(density(employee$Age),  
        col = "lightblue",  
        border = "blue")
```



14. Grouped Bar Char

```
employee <- data.frame(  
  Education = c("Bachelors", "Bachelors", "Masters", "Masters",  
                "PHD", "PHD", "Bachelors", "Masters"),  
  Gender = c("Male", "Female", "Male", "Female",  
             "Male", "Female", "Male", "Female")  
)  
group <- table(employee$Education, employee$Gender)  
barplot(group,  
  beside = TRUE,  
  col = c("skyblue", "orange", "green"),  
  legend.text = rownames(group),  
  main = "Education by Gender",  
  xlab = "Gender",  
  ylab = "Number of Employees")
```



15. Stacked Bar Chart

```
employee <- data.frame(  
  Education = c("Bachelors","Bachelors","Masters","Masters",  
    "PHD","PHD","Bachelors","Masters",  
    "PHD","Bachelors"),  
  Gender = c("Male","Female","Male","Female",  
    "Male","Female","Male","Female",  
    "Male","Female")  
)  
stack <- table(employee$Gender, employee$Education)  
barplot(stack,  
  col = c("skyblue","orange","green"),  
  legend.text = rownames(stack),  
  main = "Stacked Bar Chart of Employees",  
  xlab = "Education",  
  ylab = "Number of Employees")
```

